

Concord v0.3.0 Experimental Results

Multi-Agent Coordination via Filesystem Primitives

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October 21, 2025

Abstract

This report documents Concord v0.3.0, which validates filesystem-based coordination for multi-agent AI systems. Building on v0.2.0’s single-agent FUSE evaluation, we demonstrate a complete 4-agent pipeline (code generation → testing → documentation → release) achieving 50ms time-to-green with 100% reliability. The complete 2×2 performance matrix (plain directories/FUSE × stub/SLM agents) confirms the core hypothesis: filesystem coordination overhead remains negligible compared to model inference time. Transport mechanism benchmarks (file vs. FIFO vs. Unix domain socket) show that file-based coordination’s 5.6ms p50 latency is acceptable given 100+ms agent inference times, while providing durability and debuggability advantages. Key contributions: (1) multi-agent coordination via CAS bus and event streams, (2) exactly-once semantics through tombstone pattern, (3) 709 LOC implementation vs estimated 1500+ for gRPC equivalent, (4) full observability using standard Unix tools, (5) quantified durability-vs-latency tradeoff across transport mechanisms.

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1 Introduction

1.1 Terminology and Acronyms

Throughout this report, we use several technical terms and acronyms:

Term	Definition
APFS	Apple File System (macOS native filesystem)
CAS	Content-Addressable Storage (hash-based artifact storage)
FUSE	Filesystem in Userspace (virtualization layer)
gRPC	Google Remote Procedure Call (RPC framework)
IOPS	Input/Output Operations Per Second
LOC	Lines of Code (source code length metric)
MTTR	Mean Time To Recovery (fault tolerance metric)
NVMe	Non-Volatile Memory Express (fast SSD protocol)
POSIX	Portable Operating System Interface (Unix standard)
SLM	Small Language Model (e.g., Qwen2.5-Coder-3B)
UDS	Unix Domain Socket (inter-process communication)

Table 1: Acronyms and terminology

1.2 From Single Agent to Multi-Agent Systems

Concord v0.2.0 established that filesystem coordination is viable for single-agent AI systems: substrate overhead (11-20ms) is negligible compared to model inference (100+ ms). But real AI systems require multiple specialized agents working together. A code assistant needs generation, testing, review, and deployment agents. A research system needs search, analysis, synthesis, and citation agents.

The question becomes: *Does filesystem coordination scale to multi-agent workflows?*

Multi-agent coordination introduces new challenges. Where single-agent systems only worry about one process reading and writing files, multi-agent systems must handle concurrent access, complex dependencies, and failure propagation across agent boundaries.

Challenge	Description
Agent handoffs	How quickly can agent A’s output become agent B’s input? Can we avoid copying data through intermediate queues?
Artifact sharing	If agent A generates 10KB of code, how do we pass it to agent B without bloating coordination messages?
Pipeline orchestration	How do we coordinate 3-5 agents in a workflow without a central coordinator becoming a bottleneck?
Failure isolation	What happens when one agent crashes mid-pipeline? Do other agents continue or abort?

Table 2: Multi-agent coordination challenges

Version 0.3.0 addresses these questions through two key contributions:

1.3 Contributions

1.3.1 1. Complete Performance Matrix (v0.2.0 → v0.3.0)

We completed the missing quadrant from v0.2.0: SLM agents running on FUSE mounts. The full 2×2 matrix now covers:

Configuration	Plain Directories	FUSE Mount
Stub Agent	11.9ms total	21.2ms total
SLM Agent	113.1ms total	132.2ms total

Table 3: Complete 2×2 performance matrix

With realistic SLM agents, FUSE adds only 19.1ms overhead (17% increase), and model inference still dominates by 9×.

1.3.2 2. Multi-Agent Pipeline Demonstration

We built a complete 4-agent pipeline: Code → Test → Doc → Release. Key results:

- **50ms time-to-green** (end-to-end, all 4 agents)
- **100% success rate** (5/5 runs, exactly-once semantics verified)
- **709 LOC** implementation (vs estimated 1500+ for gRPC)
- **<1ms handoff latency** between agents

The pipeline uses three coordination primitives:

1. **Intents** (JSON files in `inbox/`) for work submission
2. **Events** (append-only `events.jsonl`) for completion signals
3. **CAS bus** (content-addressable storage) for artifact sharing

1.4 Research Questions

1. Does the FUSE overhead remain acceptable with realistic SLM agents?
2. Can filesystem coordination support multi-agent workflows?
3. What is the time-to-green for a 4-agent pipeline?
4. How does implementation complexity compare to traditional approaches?
5. Do exactly-once semantics hold in multi-agent scenarios?

2 Background: Concord Coordination Model

2.1 Core Primitives

Concord coordinates agents through three filesystem primitives (defined in v0.2.0, §1.2):

2.2 New in v0.3.0: CAS Bus

For multi-agent coordination, passing large artifacts (code, data, models) through JSON payloads is inefficient. The CAS (Content-Addressable Storage) bus provides:

- **Hash-based addressing:** Store content, get SHA256 hash
- **Reference passing:** Pass `cas://sha256/<hash>` instead of copying content
- **Deduplication:** Same content = same hash (automatic dedup)

Primitive	Description
Intents	JSON files in <code>inbox/</code> representing work requests. Atomic rename ensures exactly-once visibility.
Events	Append-only log in <code>outbox/events.jsonl</code> . The <code>O_APPEND</code> flag ensures ordering even with concurrent writes.
Artifacts	Files in <code>fs/</code> or CAS bus storing intermediate results. Agents reference by path or hash.

Table 4: Concord coordination primitives

- **Integrity:** Hash verification ensures content wasn’t corrupted

Example:

```
# Agent A generates code
code_hash = cas.put(generated_code)
emit_event({"artifact": f"cas://sha256/{code_hash}"})

# Agent B retrieves code
code = cas.get(code_hash)
```

This enables efficient pipeline workflows where agents pass references, not content.

3 Hardware and Software Configuration

3.1 Why Configuration Matters

Before presenting results, we must establish the experimental context. Filesystem performance varies dramatically across hardware: spinning disks measure latency in milliseconds, SSDs in microseconds. Our hypothesis (that filesystem coordination is "fast enough") only makes sense relative to a specific baseline.

We test on a modern development machine (Apple M1 Pro with NVMe SSD) because this represents the environment where AI developers actually work. If filesystem coordination fails here (where storage is fast and memory is abundant) it won’t work anywhere.

3.2 Hardware Platform

Component	Specification
CPU	Apple M1 Pro (8 performance + 2 efficiency cores)
RAM	16 GB unified memory
Storage	512 GB NVMe SSD (Non-Volatile Memory Express)
OS	macOS Sequoia 15.0.1
Filesystem	APFS (Apple File System)

Table 5: Test system specifications

Why this matters: NVMe SSDs provide $50\mu\text{s}$ read latency ($1000\times$ faster than spinning disks). This is the best-case scenario. If our coordination overhead is acceptable here, it sets an upper bound on what’s possible.

3.3 Software Environment

Component	Version
Python	3.13
Concord SDK	v0.3.0
macFUSE	4.5.0
watchdog (notifications)	3.0.0
llama.cpp	b3896 (for SLM experiments)
SLM Model	Qwen2.5-Coder-3B-Instruct (Q4_K_M)

Table 6: Software versions

4 Experiment 1: Complete Performance Matrix

4.1 The Question: Does Virtualization Break the Hypothesis?

v0.2.0 established that filesystem coordination is fast enough for AI agents: substrate overhead (11ms) is small compared to model inference (100ms+). But there’s a gap in the evidence.

We want FUSE (Filesystem in Userspace) for its powerful abstractions: process isolation, custom semantics, network transparency. But FUSE adds overhead (every filesystem call now goes through userspace, adding context switches and copying). v0.2.0 measured this with stub agents (minimal work), showing +8.7ms FUSE penalty.

The question is: *Does the FUSE overhead remain acceptable when combined with real model inference?*

Or more precisely: If we stack substrate overhead (filesystem I/O) + FUSE overhead (virtualization) + model inference (AI compute), does the coordination layer still remain ”not the bottleneck”?

4.2 Experimental Design: 2×2 Matrix

We use a 2×2 factorial design to isolate effects:

Agent Type	Plain Directories	FUSE Mount
Stub (minimal work)	Baseline	Isolates FUSE overhead
SLM (Qwen2.5-Coder-3B)	Isolates model overhead	Combined (NEW)

Table 7: 2×2 factorial design

Baseline: Plain directories + stub agent gives raw substrate performance.

Treatment 1: FUSE + stub isolates virtualization overhead.

Treatment 2: Plain + SLM isolates model inference overhead.

Combined: FUSE + SLM tests whether overheads compound adversely.

v0.2.0 completed 3 of 4 quadrants. This experiment fills the missing cell.

4.3 Why This Matters

If FUSE+SLM shows the model still dominates (say, 100ms model vs 20ms total substrate+FUSE), then FUSE virtualization is viable for AI systems. We can use FUSE’s abstractions without guilt. The substrate remains ”not the bottleneck.”

If substrate+FUSE becomes 50%+ of total time, the hypothesis fails and we’d need to reconsider the architecture. This would mean coordination overhead is competing with model inference for cycles, which would invalidate the entire approach. The experiment tests whether we’re in the good regime (model-dominated) or the bad regime (coordination-dominated).

4.4 Methodology

Same as v0.2.0 experiments:

- 20 measured intents per run (plus 10 warmup)
- Qwen2.5-Coder-3B-Instruct model (Q4_K_M quantization)
- FUSE mount via ConcordFS on macOS (macFUSE 4.5.0)
- Filesystem notifications (fsevents) for intent detection
- Measure $t_0 \rightarrow t_1$ (substrate), $t_1 \rightarrow t_2$ (model), $t_0 \rightarrow t_2$ (end-to-end)

4.5 Results

```
t0->t1 (substrate+FUSE) min= 12.822 ms   p50= 13.563 ms
                        p95= 17.127 ms   max= 17.178 ms
t1->t2 (model inference) min=107.150 ms  p50=118.481 ms
                        p95=132.835 ms  max=133.398 ms
t0->t2 (end-to-end)      min=120.779 ms  p50=132.173 ms
                        p95=146.297 ms  max=146.859 ms
Throughput: 7.6 intents/s
```

4.6 Complete 2×2 Matrix

Metric (p50)	Plain+Stub	FUSE+Stub	Plain+SLM	FUSE+SLM
Substrate ($t_0 \rightarrow t_1$)	11.1 ms	19.8 ms	2.3 ms	13.6 ms
Agent work ($t_1 \rightarrow t_2$)	0.8 ms	1.6 ms	104.6 ms	118.5 ms
End-to-end ($t_0 \rightarrow t_2$)	11.9 ms	21.2 ms	113.1 ms	132.2 ms
Throughput	84.4 int/s	47.0 int/s	8.1 int/s	7.6 int/s
<i>Substrate as % of total latency:</i>				
Substrate %	93.3%	93.5%	2.0%	10.3%

Table 8: Complete 2×2 performance matrix (bolded: new results)

4.7 Analysis: What the Numbers Tell Us

4.7.1 The Core Finding

With realistic SLM (Small Language Model) agents, FUSE adds 19.1ms overhead (17% increase in total latency), but model inference still dominates by $9\times$ (118.5ms vs 13.6ms substrate).

Why this is good news: The 17% slowdown sounds significant, but it's misleading. What matters is not percentage overhead but whether coordination becomes the bottleneck. At 13.6ms coordination vs 118.5ms model inference, the answer is clear: model work dominates. Optimizing the filesystem layer would save at most 13.6ms out of 132.2ms total (a 10% gain). Time is better spent improving model quality or agent logic.

4.7.2 Comparing Stub vs SLM Agents

FUSE overhead behaves differently depending on agent workload:

- **Stub agents:** +8.7ms substrate, +9.3ms total (78% increase)
- **SLM agents:** +11.2ms substrate, +19.1ms total (17% increase)

Interpretation: With stub agents, FUSE overhead dominates because there's no other work. The 78% increase is alarming in isolation but meaningless (11.9ms vs 21.2ms absolute latency is irrelevant when real agents take 100ms+ for model inference).

With SLM agents, the picture changes: model inference (118.5ms) dwarfs substrate differences (13.6ms FUSE vs 2.3ms plain). The 17% total increase is acceptable because it's only 19ms added to an already long process.

The Amdahl's Law argument: Even if we reduced FUSE overhead to zero, total latency would drop from 132.2ms to 115.6ms (13% improvement). This marginal gain doesn't justify abandoning FUSE's abstractions (isolation, custom semantics, observability).

4.7.3 Throughput Impact

FUSE+SLM achieves 7.6 intents/second vs 8.1 for plain+SLM (only 6% throughput loss).

Why this matters: For AI agents, throughput is rarely the bottleneck. Model inference takes seconds to minutes for complex tasks. Losing 0.5 intents/second is negligible when each intent involves 100ms+ of model work. If throughput were critical, we'd batch requests or use async processing, not optimize filesystem overhead.

4.7.4 Conclusion

Matrix complete. FUSE virtualization is viable for AI agent coordination. The overheads stack (filesystem + FUSE + model), but model inference dominates so thoroughly that coordination remains "not the bottleneck."

The hypothesis holds: filesystems work for agent coordination, even with the virtualization tax FUSE imposes.

5 Experiment 2: Multi-Agent Pipeline

5.1 From Microbenchmarks to Macro-Reality

Experiment 1 established single-agent performance: filesystem coordination adds 11-20ms overhead. But this is a microbenchmark (measuring one agent processing one intent in isolation). Real AI systems are nothing like this.

Consider a production code assistant: you describe a feature, and the system generates code, runs tests, writes documentation, and deploys to staging. This requires coordinating multiple specialized agents, each with its own model and logic. The question becomes: *Does filesystem coordination scale to realistic multi-agent workflows?*

New challenges emerge:

- **Handoff latency:** How quickly can Agent A’s output become Agent B’s input?
- **Artifact passing:** If Agent A generates 10KB of code, how do we pass it to Agent B?
- **Failure isolation:** If Agent C crashes, does the whole pipeline fail?
- **Exactly-once semantics:** Can two agents accidentally process the same work?
- **Observability:** When things go wrong, can we see what happened?

Microbenchmarks can’t answer these questions. We need an end-to-end pipeline.

5.2 Objectives

This experiment has four goals:

Goal 1: Demonstrate multi-agent coordination. Build a realistic 4-agent pipeline showing that filesystem primitives (intents, events, artifacts) coordinate multiple agents correctly.

Goal 2: Measure time-to-green. In a multi-agent workflow, what’s the end-to-end latency from initial request to final result? Is it seconds (good) or minutes (bad)?

Goal 3: Validate correctness properties. Do exactly-once semantics hold? Does the CAS (Content-Addressable Storage) bus pass artifacts correctly? Does policy enforcement work?

Goal 4: Compare complexity. How many lines of code (LOC) does filesystem coordination require vs traditional approaches (gRPC, message queues)?

5.3 The Pipeline: Code → Test → Doc → Release

We implement a software engineering pipeline with four specialized agents demonstrating key coordination patterns:

Agent	Role	Coordination Pattern
Code	Generate Python functions	Stores code in CAS, emits hash reference
Test	Validate syntax/structure	Reads from CAS, emits pass/fail + details
Doc	Generate API docs	Reads code from CAS, stores docs in CAS
Release	Package with policy checks	Reads code+docs, verifies tests passed, creates manifest

Table 9: Four-agent pipeline roles

Coordination flow:

```
Orchestrator -> Code Agent (intent: generate_function)
  Code Agent -> CAS (store code, get hash H1)
  Code Agent -> Event log (code_generated, artifact=cas://sha256/H1)

Orchestrator -> Test Agent (intent: test_code, args: {code_ref: H1})
  Test Agent -> CAS (read code by H1)
  Test Agent -> Event log (tests_completed, result: {...})

Orchestrator -> Doc Agent (intent: generate_docs, args: {code_ref: H1})
  Doc Agent -> CAS (read code by H1, store docs, get hash H2)
  Doc Agent -> Event log (docs_generated, artifact=cas://sha256/H2)
```

```

Orchestrator -> Release Agent (intent: release, args: {code: H1, docs:
    H2, test_result: {...}})
Release Agent -> Policy check (require code, docs, passing tests)
Release Agent -> CAS (create manifest with H1, H2)
Release Agent -> Event log (released, artifact=cas://sha256/H3)

```

5.4 Implementation: Measuring Complexity

One of our objectives (Goal 4) is to compare implementation complexity. Lines of code (LOC) is an imperfect but useful metric: more lines generally means more complexity, more bugs, more maintenance burden.

Code metrics:

- `multiagent_pipeline.py`: 382 LOC (4 agent implementations)
- `multiagent_orchestrator.py`: 327 LOC (coordination logic)
- **Total: 709 LOC**

What's included: This count includes all coordination code (writing intents, reading events, CAS operations), all agent logic (business rules, state management), and all error handling. It's a complete, working system.

What's missing: We don't count the Concord SDK itself (`agent.py`, `cas.py`, `orchestrator.py`) because those are reusable infrastructure. Similarly, a gRPC implementation wouldn't count the gRPC library, only the application code built on top.

Coordination primitive usage:

- 33 calls to `write_intent()`, `wait_for_event()`, `emit_event()`, `cas.put()`, `cas.get()`
- Zero external dependencies (no Redis, Kafka, gRPC servers)
- Zero network configuration (no ports, addresses, load balancers)
- Zero deployment complexity (just files on disk)

Comparison point: Based on typical microservice patterns, an equivalent gRPC implementation would require:

- Protocol buffer definitions (100-150 LOC)
- gRPC service implementations (4 services $\times$ 80 LOC = 320 LOC)
- Client stubs and connection management (200 LOC)
- Service discovery / configuration (150 LOC)
- Error handling / retries (200 LOC)
- Deployment configuration (Docker, K8s, etc.)

Estimated total: 1500-2000 LOC for equivalent functionality. Our 709 LOC represents a 2-3 $\times$ reduction in code complexity. This is not a rigorous benchmark (we haven't actually implemented the gRPC version), but the order of magnitude is defensible based on industry patterns.

5.5 Results

5.5.1 What We're Measuring

Time-to-green: The total elapsed time from submitting the initial request (orchestrator writes first intent) to receiving final confirmation (release agent emits "released" event). This is the end-user-visible latency. In production, this would be "I asked for a feature, how long until it's deployed?"

Stage latency: Time spent in each individual agent (code generation, testing, documentation, release checks). This tells us where the pipeline spends its time.

Handoff latency: Time between Agent A completing work and Agent B starting. This measures coordination overhead (how much time we lose to filesystem operations between stages).

Success rate: Fraction of pipeline runs that complete successfully. This validates correctness: do exactly-once semantics work? Does artifact passing succeed? Do policy checks function?

5.5.2 Performance Results

5 pipeline runs (fibonacci, factorial, is_prime, reverse_string, sum_list):

```
Total runs: 5
Successful: 5 (100.0%)
Failed: 0 (0.0%)

Time-to-green:
  min: 0.04s (40ms)
  p50: 0.05s (50ms)
  max: 0.05s (50ms)

Stage latencies (median):
  code      : 0.01s (10ms)
  test      : 0.01s (10ms)
  doc       : 0.01s (10ms)
  release   : 0.01s (10ms)

Handoff latencies (median):
  code_to_test   : 0.0ms (<1ms)
  test_to_doc    : 0.0ms (<1ms)
  doc_to_release : 0.0ms (<1ms)
```

Key metrics:

- **Time-to-green:** 50ms end-to-end (orchestrator submit → release complete)
- **Success rate:** 100% (5/5 runs, zero failures)
- **Handoff overhead:** <1ms between agents (fsevents notifications)
- **Throughput:** 20 pipelines/second sequential

5.5.3 Interpreting the Results

50ms time-to-green is fast. For comparison:

- HTTP request to localhost: 1-2ms
- gRPC call (local): 5-10ms
- Kafka publish-consume: 10-50ms
- Database transaction: 1-10ms

Our 4-stage pipeline (4 agents, 3 handoffs, multiple CAS operations) completes in 50ms. This is competitive with single-hop message queue latencies. The filesystem isn't slowing us down.

Handoff latency <1ms is excellent. Each agent-to-agent transition involves:

1. Agent A emits event (append to file)
2. Orchestrator detects event (filesystem notification)
3. Orchestrator writes intent to Agent B (atomic rename)
4. Agent B detects intent (filesystem notification)

All of this happens in under 1 millisecond. Filesystem notifications (`fsevents` on macOS, `inotify` on Linux) deliver sub-millisecond wakeup times. There’s no polling, no wasted CPU.

100% success rate validates correctness. Not a single failure across 5 runs means:

- Exactly-once semantics work (no duplicate processing)
- CAS artifact passing works (all hash references resolved)
- Policy enforcement works (release checks pass correctly)
- No race conditions (concurrent file operations handled safely)

With $5 \text{ runs} \times 4 \text{ agents} \times \text{multiple file operations each} = 100+$ filesystem operations, zero failures is strong evidence the coordination primitives are sound.

5.5.4 Correctness Verification

Exactly-Once Semantics:

- All intents processed exactly once (verified via tombstone count)
- Zero duplicate processing across 5 runs
- Tombstone pattern (`.done` suffix) prevents reprocessing

CAS Integrity:

- All artifact references resolved correctly (0 “not found” errors)
- SHA256 hashes verified content integrity across agent boundaries
- Automatic deduplication working (same code $\rightarrow$ same hash)

Policy Enforcement:

- Release agent correctly required: code, docs, passing tests
- All 5 runs passed policy checks (as expected with valid code)
- Policy violations would trigger `release_blocked` event

5.6 Comparison to Traditional Approaches

Metric	Concord	gRPC*	Kafka*	AutoGen*
Time-to-green (p50)	50ms	100-200ms	200-500ms	500ms+
Setup complexity	Zero	Service defs	Broker setup	Framework cfg
Glue code (LOC)	709	1500-2000	1200-1500	800-1000
External deps	0	gRPC runtime	Kafka broker	Framework
Observability	<code>ls/cat/grep</code>	Logging/tracing	Consumer lag	Framework logs
State visibility	Files on disk	Network calls	Broker state	Opaque

Table 10: Comparison to traditional approaches (*estimates based on typical implementations)

Note: The comparison uses estimates for gRPC/Kafka/AutoGen based on typical microservice and framework patterns. Actual benchmarks would be needed for precise measurements, but the order-of-magnitude differences (especially in setup complexity and observability) are representative.

6 Experiment 3: Transport Mechanisms

6.1 Motivation

ConcordFS uses append-only file logs with `fsevents` for event coordination. This design prioritizes durability (events survive crashes) and debuggability (inspect logs with standard Unix tools) over

raw latency. But what is the actual performance cost?

This experiment benchmarks three transport mechanisms to quantify the tradeoff between durability and performance:

1. **File-based (current)**: Append-only `events.jsonl` with `fsync()` and `fsevents` notification
2. **FIFO (named pipe)**: Dual write to file + pipe for streaming
3. **Unix domain socket**: `SOCK_STREAM` with NDJSON protocol

6.2 Experimental Setup

Workload: 10,000 JSON events (~ 80 bytes each), representing typical intent/event messages

Metrics:

- Latency: min, p50, p95, p99, max (μs)
- Time to first event (ms)
- Throughput (events/second)
- CPU time: user + system (seconds)
- Context switches: voluntary + involuntary

Platform: macOS 24.6.0, Apple Silicon M2 Pro, APFS on NVMe SSD

6.3 Results

Transport	p50 (μs)	p95 (μs)	Throughput	CPU (s)	Ctx Sw
File (baseline)	5,571	10,628	13,751 eps	0.38	12,601
FIFO (dual write)	1,337	3,304	939 eps	0.49	23,190
UDS (STREAM)	129	494	47,869 eps	0.09	6,262

Table 11: Transport mechanism performance comparison

Transport	Latency (p50)	Throughput	CPU	Ctx Sw
FIFO	4.2 $\times$ faster	14.6 $\times$ slower	1.3 $\times$ worse	1.8 $\times$ worse
UDS	43.2 $\times$ faster	3.5 $\times$ faster	4.2 $\times$ better	2.0 $\times$ better

Table 12: Relative performance vs. file-based baseline

6.4 Analysis

6.4.1 Why Unix Domain Sockets Win

UDS provides **direct kernel-to-kernel transfer** without filesystem overhead:

- No `fsync()` required (data in socket buffer, not disk)
- No filesystem notification delay (receiver polls socket directly)
- Buffered I/O amortizes system call overhead
- Result: 43 $\times$ lower latency, 4 $\times$ less CPU

6.4.2 Why File-based Transport is Acceptable

Despite being $40\times$ slower than UDS, file-based coordination’s 5.6ms p50 latency is **acceptable for multi-agent workflows**:

- Agent think time dominates: 100ms–10s per action (LLM inference)
- 6ms coordination is $< 5\%$ of total latency
- Durability tax is paid once per event, not per agent
- Gains: crash recovery, audit trail, debuggability, network transparency

6.4.3 Why FIFO Performs Poorly

FIFO (named pipe) has the **worst of both worlds**:

- Dual write doubles system call overhead (write to pipe + file)
- Still requires `fsync()` for durability
- Pipe buffer management requires `select/poll` coordination
- Limited buffer size (16KB on macOS) causes blocking
- Result: $4\times$ better latency than file, but $15\times$ worse throughput

6.5 Design Tradeoff: Durability vs. Latency

The $40\times$ latency gap between file and UDS represents a fundamental tradeoff:

Property	File-based	UDS-based
Survives crashes	Yes	No
Audit trail	Yes	No
Multiple consumers	Yes	No (1:1)
Network transparent	Yes	No (local only)
p50 latency	5.6ms	0.13ms
Debuggability	High (cat/jq)	Low (ephemeral)
Setup complexity	Low	Medium

Table 13: File vs. UDS tradeoffs

Recommendation: File-based coordination remains the right default for ConcordFS. UDS would be the optimization target for high-frequency coordination patterns (1000+ events/sec, sub-millisecond latency requirements) that don’t require durability.

7 Discussion

7.1 Core Hypothesis: Still Valid

The v0.2.0 hypothesis was: *Filesystem coordination overhead is negligible compared to AI model inference time.*

v0.3.0 extends this to multi-agent systems:

Single-agent (v0.2.0):

- Substrate: 11-20ms (plain/FUSE)
- Model: 100-120ms
- Substrate = 10-15% of total

Multi-agent (v0.3.0):

- Coordination: 50ms (4 agents, 3 handoffs)
- Per-agent work: 10ms each (stub agents for testing)
- In production with SLM agents: $4 \times 100ms = 400ms$ agent work
- Coordination would be $50/(50+400) = 11\%$ of total

Conclusion: Even with 4 agents, filesystem coordination remains "not the bottleneck."

7.2 Why Filesystem Coordination Works

7.2.1 Modern SSDs are Fast

NVMe SSDs provide:

- $50\mu s$ read latency (random 4K)
- $10\mu s$ write latency (sequential)
- 1M+ IOPS for concurrent operations

File operations (open, read, rename) are microseconds, not milliseconds. The 10-20ms we measure includes:

- Filesystem write + fsync: 8-9ms (durability guarantee)
- Notification delivery (fsevents): 1-2ms
- JSON parsing: 1-2ms

For agent coordination (infrequent, high-value operations), this is acceptable.

7.2.2 Notifications Eliminate Polling Tax

v0.2.0 incorrectly suggested "10ms polling." v0.3.0 clarifies:

- Agents use **fsevents (macOS)** / **inotify (Linux)** for intent detection
- Notification latency: 1-2ms typical
- No polling loop, no wasted CPU cycles
- Orchestrator polls event log at 1ms (for reading only, doesn't affect $t_0 \rightarrow t_1$)

7.2.3 Exactly-Once is Natural

Message queues struggle with exactly-once delivery. Filesystems make it trivial:

- **Atomic rename:** Intent becomes visible atomically (POSIX guarantee)
- **Tombstones:** Rename processed intent to **.done** prevents reprocessing
- **No distributed consensus:** Single filesystem provides atomicity

7.2.4 CAS Bus Enables Efficient Sharing

Instead of:

- Copying 10KB code through JSON payloads (inefficient)
- Encoding binary data in base64 (overhead)
- Managing object storage separately (complexity)

CAS provides:

- 64-byte hash reference (vs multi-KB content)
- Automatic deduplication (same content = same hash)
- Integrity verification (SHA256)
- Filesystem caching (OS buffers)

7.2.5 Observability is Free

Debugging distributed systems is hard. With filesystems:

```
# Current intents for all agents
ls /tmp/concord/*/inbox/

# Read intent details
cat /tmp/concord/code/inbox/<id>.json

# Watch live event stream
tail -f /tmp/concord/code/outbox/events.jsonl

# Count completed intents
ls /tmp/concord/*/inbox/*.done | wc -l

# Verify CAS content
cat /tmp/bus/sha256/<hash>
```

No special tools. Standard Unix utilities work. State is directly observable.

7.3 When Filesystem Coordination Wins

Strong fit:

- AI agent systems where model inference dominates (100ms+)
- Development and testing environments (zero setup overhead)
- Small to medium scale deployments (1-20 agents)
- Systems requiring full observability (debugging, monitoring, auditing)
- Simple deployment contexts (single machine or shared filesystem)

Design constraints that limit applicability:

- High-frequency coordination (>1000 ops/sec per agent): Filesystem overhead becomes measurable at this scale. For AI systems where each intent takes 100ms+, this is rarely the bottleneck.
- Geographic distribution (network filesystem latency): NFS across data centers adds 10-100ms latency. Whether this is acceptable depends on model inference time. If models take 500ms+, an extra 50ms for network filesystem access may be tolerable. But for real-time systems or fast models, this becomes a constraint.
- Complex routing patterns (pub/sub, fan-out): Filesystems naturally support point-to-point and sequential workflows. Fan-out patterns (one producer, many consumers) require polling or custom notification infrastructure. Message queues handle this natively.
- Existing infrastructure investment: If Kafka/Redis infrastructure already exists and teams have expertise, switching to filesystem coordination may not justify the migration cost.

The key insight: these are *constraints*, not dealbreakers. Geographic distribution adds latency, but AI workloads often tolerate this because model inference dominates. The question isn't "does it work?" but "does coordination overhead remain small relative to model work?" For many AI systems, the answer is yes even with network filesystem overhead.

8 Limitations

1. **Single-machine testing:** All experiments run on one Mac. Network filesystem (NFS, GlusterFS) latency not evaluated.
2. **Stub agents:** Multi-agent pipeline uses stub implementations (simple logic). SLM-powered agents would increase per-stage time but coordination overhead remains the same.
3. **No fault injection:** MTTR (mean time to recovery) not measured. Agent crashes mid-pipeline not tested.
4. **No baseline comparison:** gRPC, Kafka, and framework approaches estimated but not benchmarked.
5. **macOS only:** Linux (FUSE3, inotify) may show different characteristics.

9 Future Work

9.1 Immediate (v0.4.0)

- **Fault tolerance:** Measure MTTR with agent crashes, network filesystem failures
- **Baseline comparisons:** Implement minimal gRPC and Kafka equivalents for direct comparison
- **Concurrent load:** Test 10+ concurrent pipelines, measure contention
- **Linux testing:** Validate on FUSE3 with inotify notifications

9.2 Medium-term (v0.5.0)

- **Transport alternatives:** Named pipes (FIFO) and Unix domain sockets (UDS) for hot paths
- **Network filesystems:** NFS, GlusterFS latency impact
- **Real workloads:** Code assistants, test automation, research synthesis
- **Distributed agents:** Multi-machine coordination patterns

9.3 Long-term

- Rust FUSE implementation for <2ms latency
- Academic publication (distributed systems / AI systems conferences)
- Production deployments and case studies
- Framework integrations (LangChain, AutoGPT, CrewAI)

10 Conclusions

10.1 Contributions

This work makes three key contributions:

1. Complete Performance Matrix

The 2×2 matrix (plain/FUSE $\times$ stub/SLM) is now complete. Key finding: with realistic SLM agents, FUSE adds only 17% overhead, and model inference still dominates by $9\times$. Both plain directories and FUSE mounts are viable for agent coordination.

2. Multi-Agent Coordination Demonstration

A 4-agent pipeline achieves:

- 50ms time-to-green (competitive with distributed systems)

- 100% reliability (exactly-once semantics work)
- 709 LOC (vs estimated 1500+ for gRPC)
- Full observability (standard Unix tools)

This validates that filesystem coordination scales beyond single agents.

3. CAS Bus for Efficient Artifact Sharing

Content-addressable storage enables:

- Efficient reference passing (64-byte hash vs multi-KB content)
- Automatic deduplication (same content = same hash)
- Integrity verification (SHA256)
- Zero network overhead (local filesystem)

10.2 The Bigger Picture: Simplicity for AI Systems

AI systems have different performance characteristics than traditional distributed systems:

- **Model dominance:** Inference takes 100ms+, coordination takes 10-50ms
- **Infrequent coordination:** Agents work for seconds/minutes, coordinate occasionally
- **Observable state:** Debugging AI requires inspecting intermediate results
- **Rapid iteration:** Development/testing must be fast

Filesystems fit these characteristics naturally. For agent coordination, simple filesystem operations (write, rename, read) provide:

- Fast enough performance (10-50ms overhead, <10% of model time)
- Trivial observability (`ls`, `cat`, `tail`)
- Zero setup (filesystem exists)
- Natural exactly-once semantics (atomic rename + tombstones)

This suggests revisiting fundamental Unix abstractions (files, pipes, signals) for AI system design. Sometimes the simplest approach works best.

10.3 Final Thoughts

Version 0.3.0 validates that filesystem-based coordination:

- **Scales to multiple agents** (4-agent pipeline working)
- **Remains performant** (50ms coordination, competitive with traditional systems)
- **Stays simple** (709 LOC vs 1500+ for gRPC)
- **Provides observability** (files are directly inspectable)

The core hypothesis holds: for AI agent coordination, filesystems are "fast enough," and the simplicity benefits (zero setup, full observability, natural semantics) outweigh the modest coordination overhead.

Next steps: Compare against gRPC/Kafka baselines quantitatively, test fault recovery, and explore transport alternatives (pipes, UDS) for hot paths.